AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XII

ANNUAL EXAMINATIONS 2021

Chemistry

Total Time: 2 hours 10 minutes

Total Marks: 65 (50-Theory & 15-Alternate to Practical)

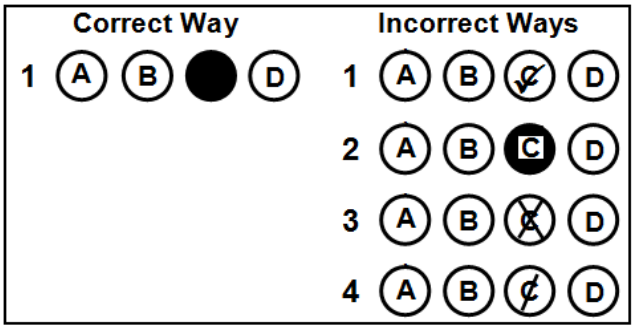
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INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 65 only.
4. Question Distribution:

Theory	Alternate to Practical (ATP)
50 MCQs	15 MCQs

5. In each question, there are four choices A, B, C, D. Choose ONE. On the answer grid, black out the circle for your choice with a pencil as shown below.



Candidate's Signature

6. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
7. DO NOT write anything in the answer grid. The computer only records what is in the circles.
8. The marks obtained on the 50 MCQs will be equated to the total marks of 85 for the theory examination results.
9. You may use a scientific calculator if you wish.

THEORY (Questions 1-50)

1. Hydrofluoric acid (HF) is classified as the weakest acid because it possesses

   A. low ionisation.
   B. weak polarisation.
   C. low bond enthalpy.
   D. weak hydrogen bonding.

2. Consider the following properties.

   I. Low electronegativity
   II. High ionisation enthalpy
   III. Small atomic and ionic size

   Amongst group IIA elements, beryllium shows anomalous behaviour MAINLY due to

   A. I only.
   B. II only.
   C. I and III.
   D. II and III.

3. An element 'indium' belongs to group IIIA and period 5 of the modern periodic table.

   The electronic configuration for the given element is

   A. $1s^2 2s^2 2p^6 3s^2 3p^3$
   B. $1s^2 2s^2 2p^6 3s^2 3p^5$
   C. $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 5s^2 4d^{10} 5p^1$
   D. $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 5s^2 4d^{10} 5p^3$

4. Thermodynamically, the MOST stable oxidation state of carbon is

   A. + 4
   B. + 2
   C. – 2
   D. – 4

5. Elements which often show variable oxidation states and form coloured compounds are called

   A. halogens.
   B. noble gases.
   C. transition metals.
   D. alkaline earth metals.

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6. If the coordination sphere carries a negative charge, then the suffix for transition metal ion in International Union of Pure and Applied Chemistry (IUPAC) nomenclature would be

   A. ate.
   B. ene.
   C. ium.
   D. ane.

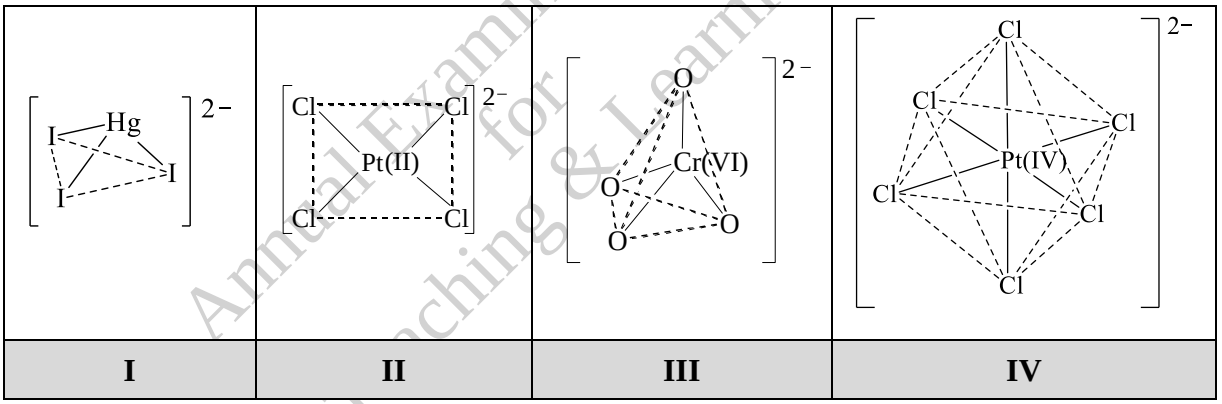
7. Consider the given reaction in aqueous medium.

   $10FeSO_4 + 2KMnO_4 + 8H_2SO_4 \rightarrow 5Fe_2(SO_4)_3 + 2MnSO_4 + K_2SO_4 + 8H_2O$

   In the given reaction, the purple colour of potassium permanganate fades away gradually because the

   A. ferrous ions are converted into ferric ions.
   B. water molecules dilute the reaction mixture.
   C. permanganate changes to manganese sulphate.
   D. dilute solution of sulphuric acid is used in the reaction.

8. Consider the following structural arrangements of ligands around the central ion.



   The structures that have the coordination number 4 are

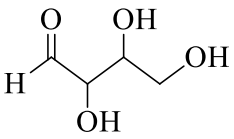
   A. I and II.
   B. I and IV.
   C. II and III.
   D. III and IV.

9. The condition required to extract aliphatic and aromatic components from coal is

   A. high pressure.
   B. varying temperature.
   C. excessive supply of air.
   D. different amounts of coal.

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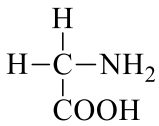
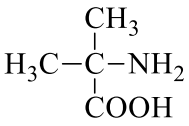
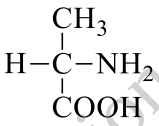
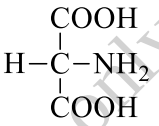
10. Consider the given structure of four-carbon sugar.



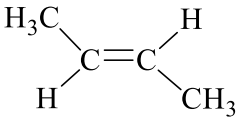
How many chiral centres are present in this structure?

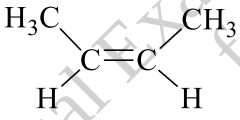
A. 1
B. 2
C. 3
D. 4

11. The compound that has an optical isomer is

			
A	B	C	D

12. Given molecules are the isomers of each other.

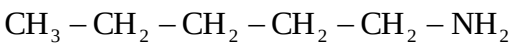




The type of isomerism these molecules exemplify is

A. chain isomerism.
B. optical isomerism.
C. functional isomerism.
D. geometrical isomerism.

13. Given is one of the positional isomers of aliphatic amine.



The number of possible positional isomers for this amine is

(**Note**: The count should not include the given structure.)

A. two.
B. three.
C. four.
D. five.

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14. The residue left behind after destructive distillation of coal is

    A. coke.
    B. coal tar.
    C. coal gas.
    D. carbon dioxide.

15. The minimum number of carbon atoms required to show chain isomerism in aldehyde is

    A. 2
    B. 3
    C. 4
    D. 5

16. An example of the simplest arene is

    A. diene.
    B. amine.
    C. ethene.
    D. benzene.

17. Consider the following structural features of a class of compounds.

    - Each carbon is sp^3 hybridised.
    - Complete rotation is restricted around the single bond.
    - Constitutional isomers are possible.
    - The most stable form contains six carbon atoms.

    The class of compounds that exhibits these features is

    A. alkanes.
    B. alkenes.
    C. cycloalkanes.
    D. substituted benzene.

18. A common reaction for both the molecules of ethene and benzene is

    A. addition reaction.
    B. elimination reaction.
    C. substitution reaction.
    D. polymerisation reaction.

19. The given equation shows a reaction of ethyne with the sodium metal.

$$HC \equiv CH + Na_{(s)} \rightarrow HC \equiv C^{-}Na^{+} + \frac{1}{2}H_{2(g)}$$

    In this reaction, the acidity of ethyne is due to a

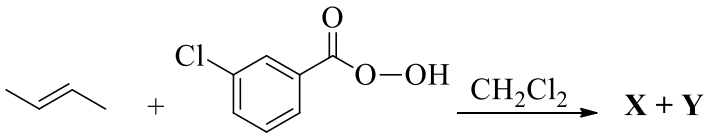
    A. high percentage of 's' character in sp hybridised carbon.
    B. stronger attraction between carbon and hydrogen.
    C. high electronegativity of terminal hydrogen.
    D. less electronegativity of terminal carbon.

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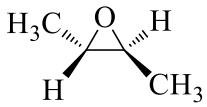
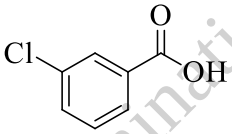
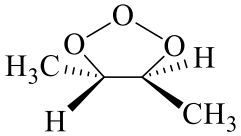
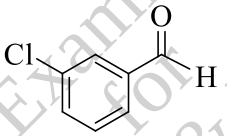
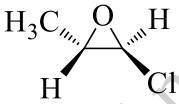
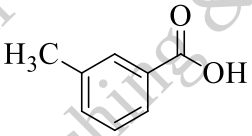
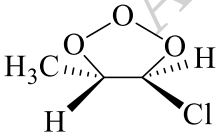
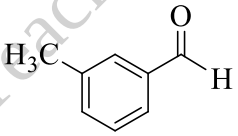
20. When ultraviolet radiations are passed through a gaseous mixture of bromine and ethane, it forms bromoethane by following the mechanism of

A. radical substitution.
B. bimolecular elimination.
C. nucleophilic substitution.
D. unimolecular elimination.

21. Consider the following reaction.



The products **X** and **Y** in the given reaction are

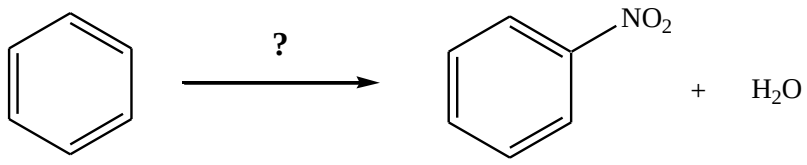
	X	**Y**
A		
B		
C		
D		

22. Which of the following acts as a meta-directing group in the electrophilic substitution reaction of benzene?

A. –Cl
B. –OH
C. –CN
D. $–N(CH_3)_2$

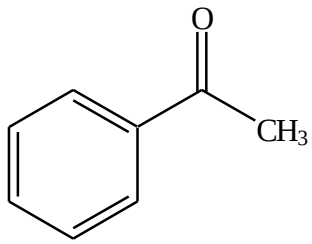
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23. The reagent(s) required for the given chemical reaction is/ are



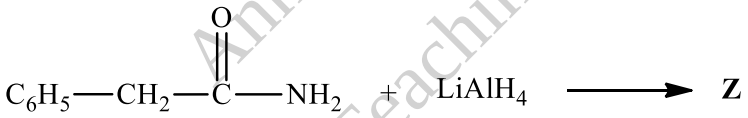
A. H_2SO_4
B. $NaNO_2$
C. HNO_3 and H_2SO_4
D. $NaNO_2$ and H_2SO_4

24. The CORRECT number of sigma bonds in the given organic compound is



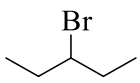
A. 13
B. 14
C. 17
D. 21

25. The product **Z** in the given chemical reaction is



A. $C_6H_5-CH_3$
B. $C_6H_5-CH_2-CH_3$
C. $C_6H_5-CH_2-NH_2$
D. $C_6H_5-CH_2-CH_2-NH_2$

26. Consider the given alkyl halide.

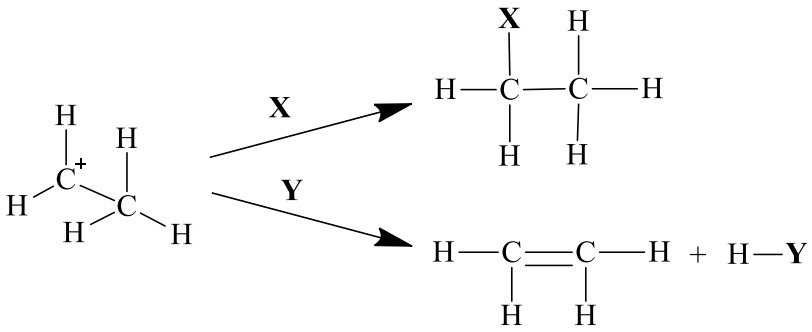


If the given alkyl halide is heated with alcoholic potassium hydroxide, then the favoured mechanism of reaction will be

A. S_N1
B. S_N2
C. E1
D. E2

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27. Given are the two ways through which a secondary carbocation can react in a polar solvent.



In the given scheme, **X** and **Y** act as

	X	**Y**
A	nucleophile	base
B	base	nucleophile
C	nucleophile	leaving group
D	base	leaving group

28. Given are the different types of alkyl halides on the basis of 'R' groups.

I. RCH_2-X
II. R_2CH-X
III. R_3C-X
IV. CH_3-X

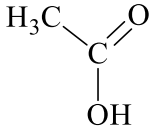
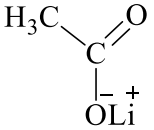
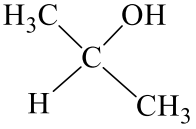
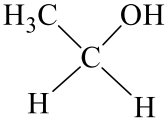
The increasing order of reactivity of these alkyl halides for the S_N2 reaction mechanism is

A. I → II → III → IV
B. III → II → I → IV
C. IV → III → II → I
D. IV → I → II → III

29. Which of the following is TRUE about diethyl ether molecule?

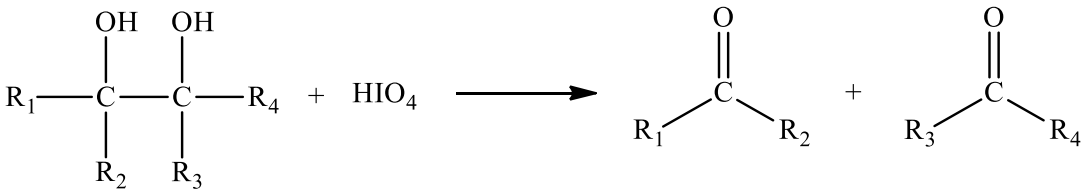
A. It is a liquid with low volatility.
B. It has zero net dipole moment.
C. It is highly miscible in water.
D. It has an angular geometry.

30. The addition of $LiAlH_4$ to ethanal followed by hydrolysis yields

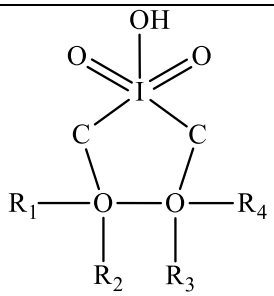
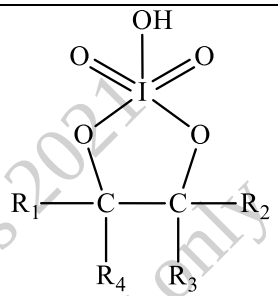
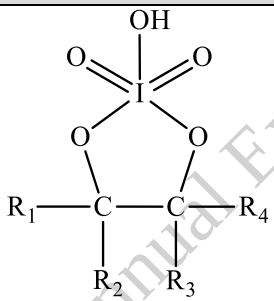
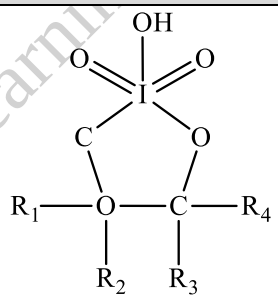
			
A	B	C	D

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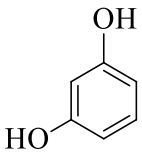
31. The oxidative cleavage of 1,2-diol by periodic acid (HIO_4) gives two carbonyl compounds. The reaction occurs via the formation of a cyclic periodate ester.



The structure of cyclic periodate intermediate formed in the given reaction would be

	
A	B
	
C	D

32. Given is the structure of resorcinol.



The International Union of Pure and Applied Chemistry (IUPAC) name of this compound is

A. o-Dihydroxyphenol.
B. p-Dihydroxyphenol.
C. 1,3-Dihydroxybenzene.
D. 1,5-Dihydroxybenzene.

33. Which of the following exemplifies the use of alcohol as a disinfectant?

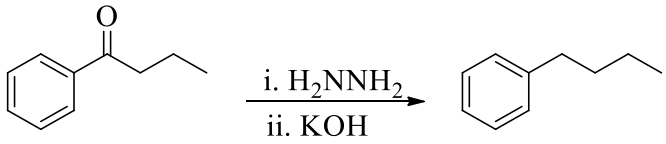
A. Cleaning skin before an operation
B. Using mouth wash for oral infections
C. Treating minor cuts, burns and wounds
D. Sterilising a needle before draining blood

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34. In contrast to alcohols, phenols are

A. less toxic.
B. less acidic.
C. used in beverages.
D. sparingly soluble in water.

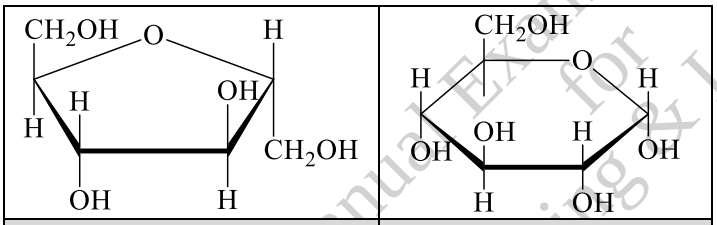
35. Consider the given reaction of ketone.



The given reduction reaction of ketone is identified as

A. nitrogen nucleophile reduction.
B. Wolff-Kishner reduction.
C. Clemensen reduction.
D. hydride reduction.

36. Consider the given closed ring structures of two different carbohydrates.



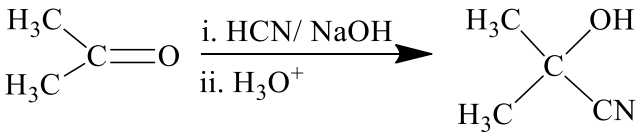
I **II**

The functional group(s) that structures **I** and **II** contain(s) in their open chain structures is/ are

	I	**II**
A	ketone	ketone
B	aldehyde	aldehyde
C	aldehyde	ketone
D	ketone	aldehyde

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37. Given is the nucleophilic addition reaction of propanone.



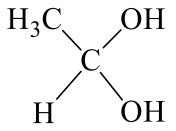
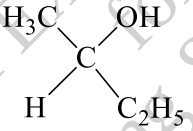
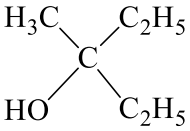
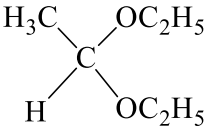
The nucleophile that attacks the carbonyl carbon in this reaction is the

A. active pair of electrons on nitrogen in CN^-.
B. active pair of electrons on carbon in CN^-.
C. OH^- ion in NaOH.
D. H^+ ion in H_3O^+.

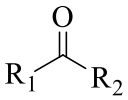
38. A colourless, strong smelling gas that contains a carbonyl group and is used in making pressed-wood products as well as a preservative of biological specimen is

A. acetone.
B. acetaldehyde.
C. acetophenone.
D. formaldehyde.

39. The major product that is obtained when one mole ethanal reacts reversibly with two moles of ethanol in the presence of dry hydrogen chloride gas is

			
A	B	C	D

40. Given is the general structure of a carbonyl compound.

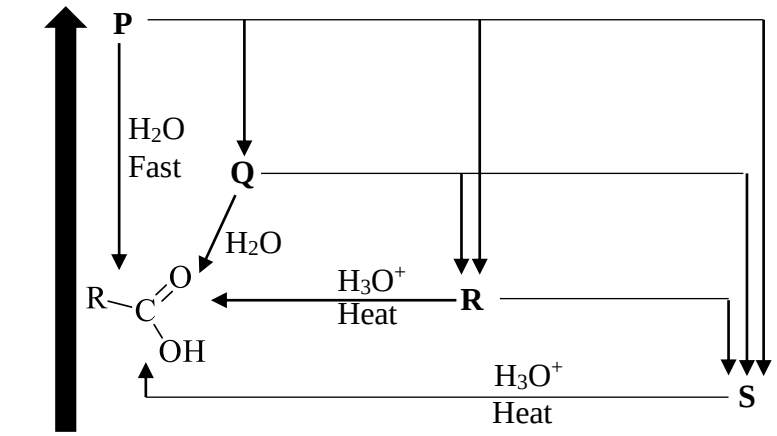


The carbonyl compound only shows iodoform reaction when **R_1** and **R_2** are

	R_1	**R_2**
A	H	CH_3
B	H	H
C	CH_2CH_3	H
D	CH_2CH_3	CH_2CH_3

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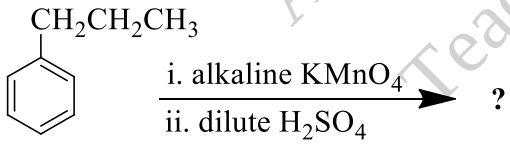
41. The given diagram shows the interconversion of carboxylic acid derivatives represented as **P**, **Q**, **R** and **S**.



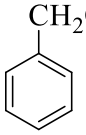
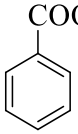
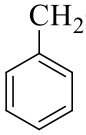
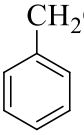
The option that CORRECTLY depicts **P**, **Q**, **R** and **S** is

	P	**Q**	**R**	**S**
A	Amide	Ester	Acid anhydride	Acyl chloride
B	Acyl chloride	Acid anhydride	Ester	Amide
C	Acid anhydride	Acyl chloride	Amide	Ester
D	Ester	Amide	Acyl chloride	Acid anhydride

42. Given is the reaction of n-propyl benzene with alkaline $KMnO_4$ followed by acidic hydrolysis.



The MAIN product obtained at the completion of this reaction will be

			
A	B	C	D

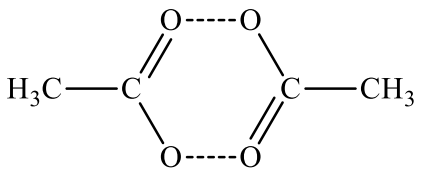
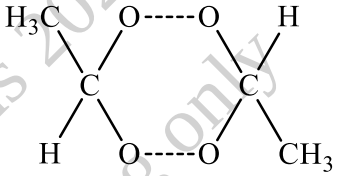
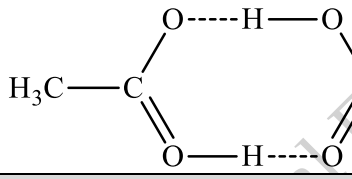
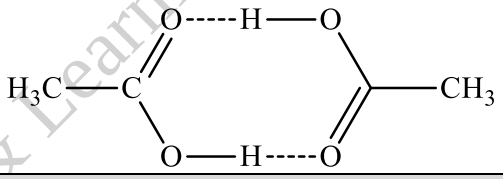
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43. Which of the following CORRECTLY represents **X** and **Y** in the given equation?

$$CH_3C{\equiv}N + 2H_2O + HCl\ (excess) \longrightarrow \mathbf{X} + \mathbf{Y}$$

	X	**Y**
A	CH_3COCH_3	NH_4Cl
B	CH_3COOH	NH_4Cl
C	CH_3COCl	NH_3
D	CH_3CH_2Cl	NH_3

44. The structure that represents the dimer of carboxylic acid is

	
A	B
	
C	D

45. Two moles of Grignard's reagent are used to convert an ester into a tertiary alcohol.

If one mole of Grignard's reagent is added, then the product formed is a/ an

A. secondary alcohol.
B. carboxylic acid.
C. aldehyde.
D. ketone.

46. Which of the following compounds is the MOST acidic?

A. F_3CCOOH
B. CH_3COOH
C. $F_2CHCOOH$
D. FCH_2COOH

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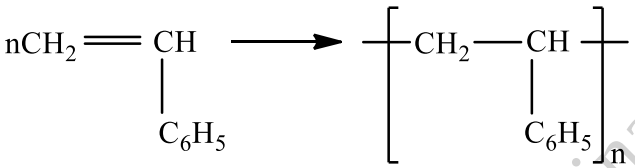
47. The dye that was prepared for the first time on fabric by the reaction of diazotised p-nitroaniline with β-naphthol was

A. Bismarck brown.
B. methyl orange.
C. indigo blue.
D. para red.

48. Which of the following fractions of petroleum is used to obtain petrol?

A. Diesel
B. Gasoline
C. Kerosene oil
D. Petroleum gas

49. The given equation shows a petrochemical and its derivative.



Which of the following pairs of monomers and polymers is shown in the given polymerisation process?

	Monomer	**Polymer**
A	Vinyl chloride	Polyvinyl chloride
B	Styrene	Polystyrene
C	Ethylene	Polyethylene
D	Butadiene	Polybutadiene

50. A class of dyes has the following features.

- It has no natural affinity towards fabric.
- It is applied to fabric with the help of salts.
- It forms an insoluble coordinate complex between the dye and the fabric.

These dyes are MOST suitable for

A. wool.
B. cotton.
C. polyester.
D. cellulose acetate.

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ALTERNATE TO PRACTICAL (ATP: Questions 51-65)

51. A student performed an experiment and recorded the observation in the given table.

Experiment	Observation	Inference
Original salt solution + a few drops of HNO_3 + boil and cool + a few crystals of NH_4Cl + NH_4OH in excess	A gelatinous mass is obtained	**X**

Based on this observation, **X** can be inferred as the salt of all of the following basic radicals EXCEPT

A. Pb^{2+}
B. Fe^{2+}
C. Cr^{3+}
D. Al^{3+}

52. In salt analysis, the basic radical that is detected as group II sulphide is

A. Mn^{2+}
B. Cu^{2+}
C. Co^{2+}
D. Ni^{2+}

53. A student performed an experiment and recorded the observation and inference in the given table.

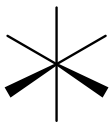
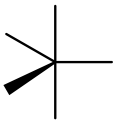
Experiment	Observation	Inference
Original salt + dilute HCl	Brisk effervescence which turned lime water milky	Gas **X** is evolved which indicates the presence of **Y** radical

Based on the given information, **X** and **Y** are

	X	Y
A	CO_2	CO_3^{2-}
B	SO_2	SO_3^{2-}
C	Cl_2	Cl^-
D	NO_2	NO_3^-

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54. The structure of complex ion in $[Cu(NH_3)_4]SO_4$ will be

			
A	B	C	D

55. The purpose to form Lassaigne's solution before element detection in an organic compound is to convert a/ an

A. solid into liquid.
B. element into its ionic form.
C. covalent bond into metallic bond.
D. soluble substance into its insoluble form.

56. To the Lassaigne's solution of an unknown organic compound, a few crystals of $FeSO_4$ along with the aqueous solution of NaOH and dilute H_2SO_4 are added. This reaction mixture is then heated.

The result of this test is a Prussian blue coloured suspension which indicates the presence of

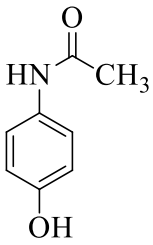
A. sulphur.
B. chlorine.
C. bromine.
D. nitrogen.

57. Treatment of **X** with HNO_2 and HCl at 0°C will produce an unstable intermediates of diazonium salt. The compound **X** is identified as the

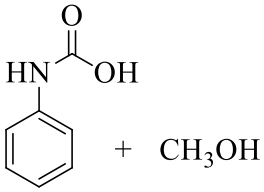
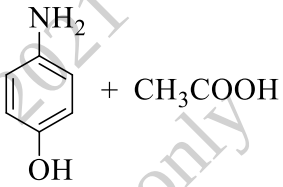
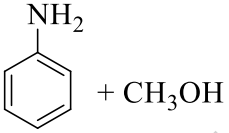
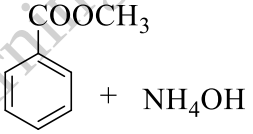
A. primary aliphatic amine.
B. primary aromatic amine.
C. secondary aliphatic amine.
D. secondary aromatic amine.

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58. Given is the structure of a pain killer drug, paracetamol.



When this structure is hydrolysed under acidic condition, it produces

Product	
	
A	B
	
C	D

59. The characteristic odour that indicates the formation of iodoform crystals is

A. ethereal.
B. pungent.
C. analogous to chloroform.
D. analogous to rotten eggs.

60. Consider the given physical properties of an aromatic class of compounds.

- They have a specific antiseptic odour.
- They are colourless in pure form.
- They may exist in a solid or liquid state.
- They become coloured upon air oxidation.
- They have a high boiling point due to hydrogen bonding.

Based on these properties, the aromatic class of compounds identified is

A. ketones.
B. amines.
C. phenols.
D. aldehydes.

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61. Oxidation of aldehydes with alkaline $KMnO_4$ results in corresponding

A. ethers.
B. ketones.
C. alcohols.
D. carboxylic acids.

62. Upon being subjected to a silver mirror test, an unknown organic compound has shown a positive result.

Which of the following inference can be drawn from this observation?

A. An aldehyde is oxidised to carboxylic acid and silver ions are reduced to silver metal.
B. An aldehyde is reduced to primary alcohol and silver metal is oxidised to silver ions.
C. A ketone is reduced to secondary alcohol and silver metal is oxidised to silver ions.
D. A ketone is oxidised to carboxylic acid and silver ions are reduced to silver metal.

63. A student was given an unknown compound to perform a preliminary test. During the test, the student observed that it smelled like vanilla.

Based on this observation, it can be inferred that the compound may contain a/ an

A. ketone.
B. protein.
C. aldehyde.
D. carbohydrate.

64. In the presence of concentrated H_2SO_4, when a carboxylic acid is heated with C_2H_5OH, it forms **X** which is indicated by **Y**.

This **X** and **Y** are identified as

	X	**Y**
A	ester	fruity smell
B	aldehyde	bitter almond odour
C	ether	buff coloured precipitate
D	amine	orange coloured dye

65. When a sample of carboxylic acid is heated with a mixture of NaOH and $Ca(OH)_2$, a burning soot is observed.

This observation inferences that the carboxylic acid present in the sample is

A. acetic acid.
B. oxalic acid.
C. formic acid.
D. benzoic acid.

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